

One Page Report				
Name of the Course		Make – A – Product: Phase I&II		Regulation
MAP – Title		Design & Development of Smart Automated Broom Stick Cleaning Machine		
Team Members (Regular(R) / Optional(O)):				
S.No	Register No.	Name of the Student(s)	R/O	Year/Dept./Section
1	611223114080	SABARI K	R	III/MECH/B
2	611223114064	NAVEEN KUMAR S	R	III/MECH/B
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Problem Statement		Manual broom stick cleaning and husk removal is a labour-intensive and time-consuming process in small-scale industries. Each worker can clean only around 80–100 broom sticks per day, taking approximately 2 - 3 minutes per stick. This results in low productivity, inconsistent cleaning quality, and health issues due to dust exposure. Additionally, industries rely heavily on multiple labourers, increasing operational cost.		
Target Customer		The primary target customers are small and medium-scale broom manufacturing units, especially in rural and semi-urban areas. These industries require affordable, easy-to-use, and low-maintenance automation solutions to improve productivity and reduce manual effort.		
Solution Methodology&Product		The proposed solution is the Design and Development of a Smart Automated Broom Stick Cleaning Machine . The system uses a combination of rotating nylon brushes, roller feeding mechanism, and a suction blower for effective husk removal. A 12V DC motor drives the cleaning mechanism. The machine is designed to be compact, portable, and user-friendly with simple ON/OFF control.		
Uniqueness of Product (Max. 5 Points)		• Low-cost automation suitable for small industries (below ₹5000) • Compact and portable design for easy handling • Reduces manual labour and improves productivity • Integrated dust collection system for safer operation • Simple design with low maintenance and easy operation		
Snapshot		Product photo 	Team photo 	
SDG No(s). Alignment	Decent Work and Economic Growth	Technical Club Aligned	Agriculture and Automaion	
MAP Guide Name Designation	MR. J. PRAKASH ASSISTANT PROFESSOR MECH	MAP Coordinator Name Designation	MR. A GAJENDRAN M.E ASSISTANT PROFESSOR MECH	
Signature		Signature		